

Tennis Video Analysis: Ball Tracking, Court Detection & Player Localization

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Abstract

We present three independent deep learning pipelines for spatial analysis of broadcast tennis video. Ball tracking: TrackNet, a fully-convolutional encoder-decoder taking 3 stacked RGB frames (9-ch, 640×360) and regressing a 256-class Gaussian heatmap with focal loss (F1 = 0.927, 96.4% of detections within 5 px). Court detection: identical backbone re-headed to 15 keypoint heatmaps—partial convergence after lr correction (F1 = 0.129, 12.9% keypoint accuracy). Player localization: fine-tuned YOLOv8s (mAP@0.5 = 0.978, mAP@0.5:0.95 = 0.711). All models trained on the Alan HPC cluster and tracked via Weights & Biases.

Problem Statement

Classical tracking fails under broadcast tennis conditions. Key challenges:

- Ball: only 2–5 px wide at 640×360; high speed causes motion blur; frequently occluded by players or net.
- Court: 14 boundary keypoints must be located simultaneously under perspective distortion and camera motion.
- Players: large scale variation, mutual occlusion, cluttered advertising backgrounds.

Methodology

Datasets

Task	Dataset	Frames	Split
Ball	TrackNet Kaggle (10 games, ≤15 clips/game)	19 800	70/15/15%
Court	GitHub keypoint dataset (JSON, 14 kp + centre)	9 000	70/15/15%
Player	Roboflow tennis-player (1 class, bbox)	5200	80/10/10%

Ball Tracking — TrackNet

Ball Tracking — TrackNet

Three consecutive RGB frames stacked → 9-ch 640×360 input, giving the network explicit motion context. Fully-sequential VGG encoder-decoder (no skip connections): 3 MaxPool stages halve spatial resolution to 80×45; 3 Upsample stages restore it.

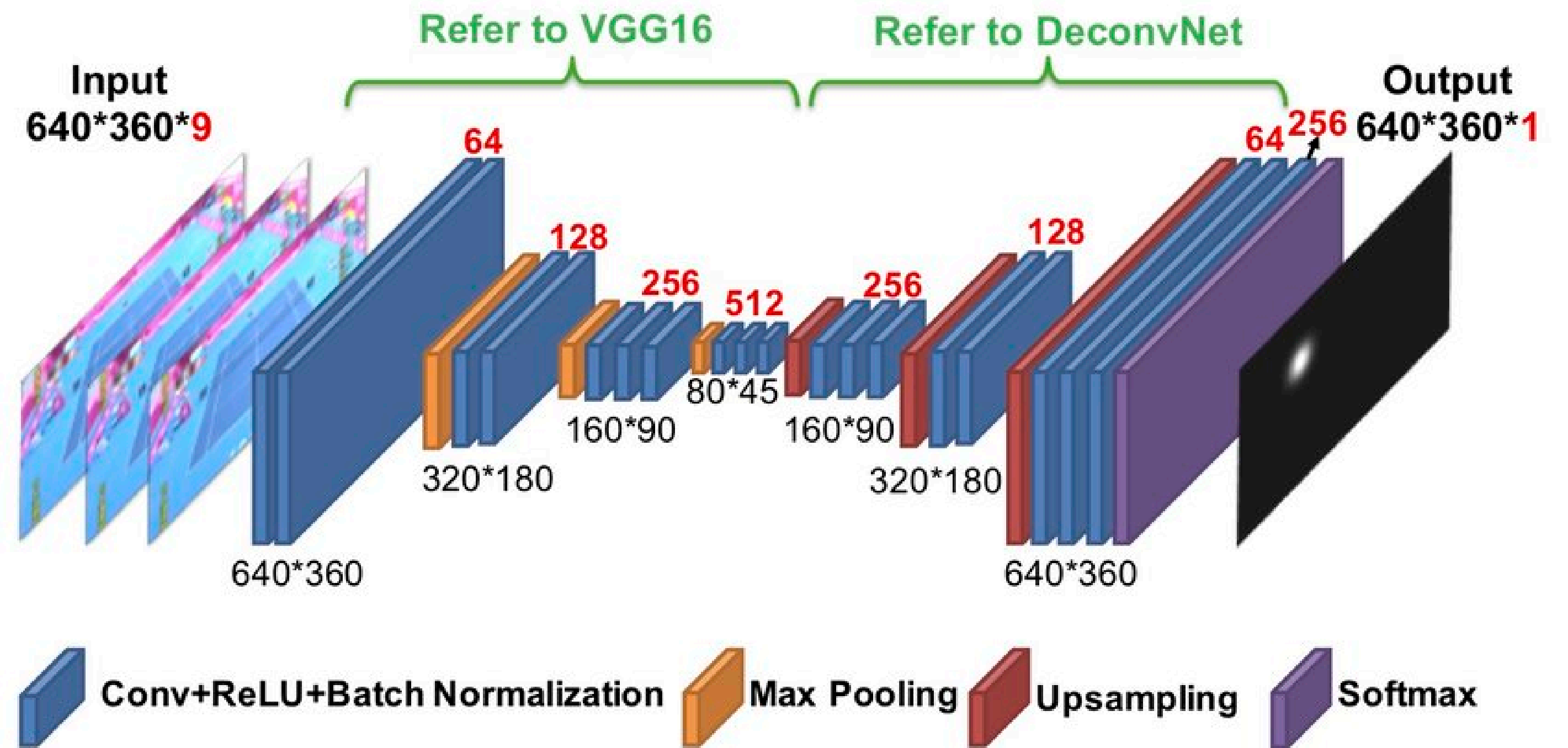


Figure 1. TrackNet: sequential encoder-decoder. Block = Conv2d–BN–ReLU. MP=MaxPool, US=Upsample (×2). Kaiming init, Dropout2d p=0.3.

Heatmap target — Gaussian centred on ball, discretised to 256 intensity classes ($\sigma^2=10$):

$$y_{h,w} = \left\lfloor 255 \cdot e^{-\|(h,w)-\mathbf{p}^*\|^2/20} \right\rfloor \quad (1)$$

Focal loss ($\gamma=2$) down-weights the dominant background class:

$$\mathcal{L} = -(1/N) \sum_i (1 - \hat{p}_i)^2 \log \hat{p}_i \quad (2)$$

Training: Adam · lr=5×10^{−4} · batch 4 · 30 epochs · fp16 via HuggingFace Accelerate · grad clip ||g||≤1.0 · dropout p=0.3 · ImageNet normalisation

Detection rule: max_{*h,w*} $\hat{y} > 2.55$. TP if predicted centroid within 5 px of ground truth; TN if ball absent and suppressed.

Court Detection — TrackNetCourt

Court Detection — TrackNetCourt

Identical encoder-decoder backbone as TrackNet; output head replaced with 15 heatmaps (14 court boundary keypoints + centre, via diagonal intersection). Single RGB frame at 320×176. Weighted MSE loss (pos_weight = 1000) to compensate for sparse keypoint activations (<0.01% of pixels active per heatmap).

Player Tracking — YOLOv8s

Player Tracking — YOLOv8s

Pretrained YOLOv8s fine-tuned end-to-end, single class tennis-player, multi-scale CIoU bounding-box regression. Evaluated with COCO mAP@0.5 and mAP@0.5:0.95.

Results

Ball Tracking

F1 0.927	Precision 0.961	Recall 0.895	Accuracy 0.868
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TP 3137	FP 126	FN 370	TN 117
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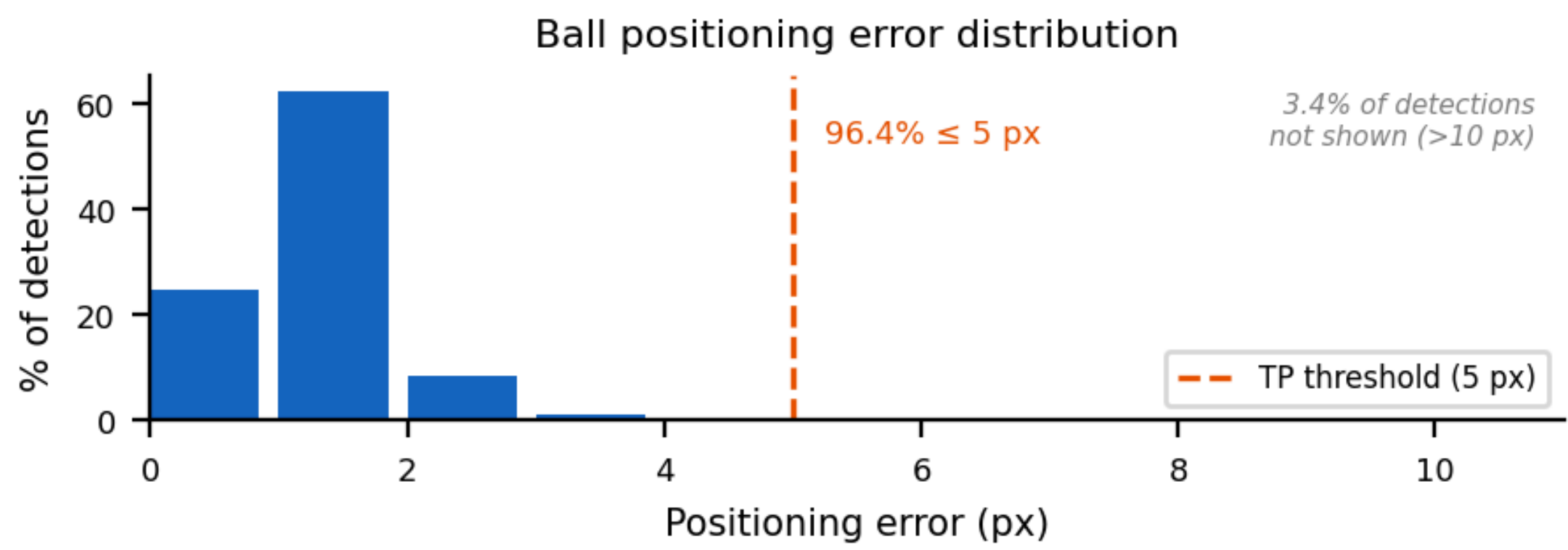


Figure 2. Positioning error on 3 253 detected balls. Blue = TP (< 5 px); grey = FP. 3.4% > 10 px not shown.

Court Detection

TP / total 3 845 / 29 835	Recall 12.9%	F1 0.129
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Partial convergence at lr = 10^{−5} (vs. 10^{−3} in the first run, which gave TP = 16). Remaining bottlenecks: weighted MSE with pos_weight = 1000 is unstable when positive signal covers under 0.01% of pixels; 320×176 resolution limits spatial precision at the 7 px threshold.

Player Tracking

mAP@0.5 0.978	mAP@0.5:0.95 0.711	F1 0.957
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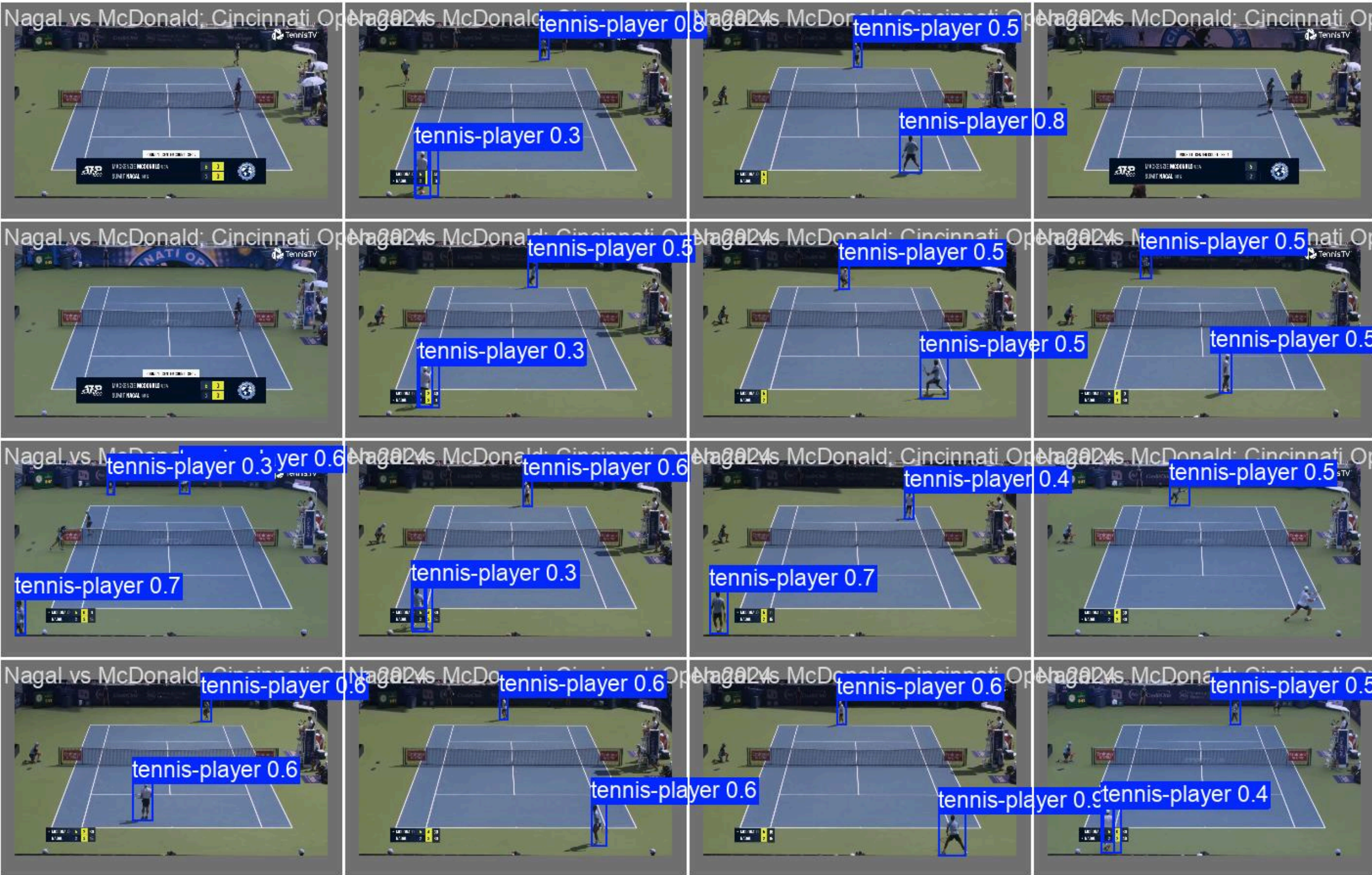


Figure 3. YOLOv8s detections on a validation batch. Per-box confidence scores shown for class tennis-player.

Conclusion

TrackNet achieves F1 = 0.927 with 96.4% of detections within 5 px, validating the 3-frame motion-context design. YOLOv8s fine-tuning reaches mAP@0.5 = 0.978 with minimal task-specific engineering. Court keypoint regression failed to converge—lr schedule and focal heatmap loss are the primary remediation targets.

Future Work

- Court detection: lower lr to 10^{−3}, switch to focal heatmap loss, increase resolution.
- Ball tracking: trajectory smoothing, bounce detection.
- Unified pipeline: fuse all three outputs into a spatiotemporal analyser.
- Datasets: larger/more diverse broadcast video; semi-supervised annotation.

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